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Endoscope and endoscope system

Abstract

An endoscope comprising an insertion section including a distal end portion, a distal end opening in a distal end surface of a distal end portion, a side opening in a side surface of the distal end portion, a suction channel located in the insertion section and extending proximally from the distal end portion, and a valve switchable between an open state and a closed state. When a first suctioning flow through the distal end opening is less than a minimum suctioning flow, the valve is in the open state and a second suctioning flow through the side opening is present, and when the first suctioning flow is more than the minimum suctioning flow, the valve is in the closed state and the second suctioning flow is absent.

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References Cited

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
368325	12/1948	JP	N/A
H10-179510	12/1997	JP	N/A

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Background/Summary

RELATED APPLICATION DATA (1) This application is based on and claims priority under 35 U.S.C. § 119 to U.S. Provisional Application No. 63/320,772 filed on Mar. 17, 2022, the entire contents of which are incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates to an endoscope including a suction function, and an endoscope unit.

BACKGROUND

(2) In examination and treatment using an endoscope, blood and the like have to be suctioned to identify a bleeding part at a time when there is bleeding inside a subject. For example, Japanese Patent Application Laid-Open Publication No. H10-179510 discloses a technique with which suction inside a body cavity or the like may be efficiently performed in a case where a distal end of a passage for suction provided in a distal end rigid portion is opened from a distal end surface of an insulating cap along a side surface and is used as a suction passage.

SUMMARY OF THE DISCLOSURE

(3) An endoscope according to an aspect of the present disclosure includes: an insertion section including a distal end portion; a distal end opening in a distal end surface of the distal end portion; a side opening in a side surface of the distal end portion; a suction channel located in the insertion section and extending proximally from the distal end portion, wherein the suction channel is in communication with the distal end opening and with the side opening; and a valve switchable between an open state and a closed state, wherein when a first suctioning flow through the distal end opening is less than a minimum suctioning flow, the valve is in the open state and a second suctioning flow through the side opening is present, and wherein, when the first suctioning flow through the distal end opening is more than the minimum suctioning flow, the valve is in the closed state and the second suctioning flow through the side opening is absent.

(4) An endoscope system according to an aspect of the present disclosure includes: an endoscope, wherein the endoscope includes: an insertion section including a distal end portion, a first opening in a distal end surface of the distal end portion, a second opening in a side surface of the distal end portion; and a suction channel located in the insertion section and extending proximally from the distal end portion, wherein the suction channel is in communication with the distal end opening and with the side opening, a distal cover attached to the distal end portion, wherein the distal cover includes: a valve switchable between open state and closed state, wherein, when a first suctioning flow through the distal end opening is less than a minimum suctioning flow, the valve is in the open state and a second suctioning air flow through the side opening is present, and wherein, when the first suctioning flow through the distal end opening is more than the minimum suctioning flow, the valve is in the closed state and the second suctioning flow through the side opening is absent.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram schematically showing an endoscope system of an aspect of the present disclosure.
- (2) FIG. 2 is a perspective view showing a configuration of a distal end part of an insertion section to which a cover member is attached.
- (3) FIG. 3 is a cross-sectional view showing the configuration of the distal end part of the insertion section to which the cover member is attached.
- (4) FIG. 4 is an exploded perspective view showing the cover member, and the distal end part of the insertion section before the cover member is attached.
- (5) FIG. 5 is a perspective view showing a configuration of the cover member.
- (6) FIG. 6 is a plan view showing the configuration of the cover member where an open/close valve is formed by two slits.
- (7) FIG. 7 is a cross-sectional view showing a state where a distal end portion is in contact with a body cavity wall and a distal end opening portion is blocked.
- (8) FIG. 8 is a perspective view showing a configuration of the distal end part of the insertion section to which the cover member of a first modification is attached.
- (9) FIG. 9 is a plan view according to a second modification, showing the configuration of the cover member where three slits are formed.
- (10) FIG. 10 is a plan view according to a third modification, showing the configuration of the cover member where the open/close valve having a small width on a distal end side is formed by two slits.
- (11) FIG. 11 is a plan view according to a fourth modification, showing the configuration of the cover member where the open/close valve having a great width on the distal end side is formed by two slits.
- (12) FIG. 12 is a cross-sectional view according to a fifth modification, showing the configuration of the distal end part of the insertion section where an open/close valve is provided at a distal end structural part.
- (13) FIG. 13 is a cross-sectional view according to the fifth modification, showing a state where the distal end portion is in contact with a body cavity wall and the distal end opening portion is blocked.

DETAILED DESCRIPTION

(14) Hereinafter, a configuration of an aspect of the present disclosure will be described. Note that in the following description, the drawings based on the embodiment are schematic, and a relationship between a thickness and a width of each part, a ratio of respective thicknesses of parts, and the like are different from actual relationships. Moreover, dimensional relationships, ratios and

the like may be different between drawings.

(15) At a time of performing suction inside a subject using an endoscope, a body cavity wall or the like may stick to a distal end portion of an insertion section to completely block a channel opening or to reduce an opening amount. Particularly, when the body cavity wall or the like sticks to the channel opening at a time of performing suction of blood or the like by the endoscope to treat bleeding, suction of blood or the like cannot be continued.

(16) In such a case, a suction operation by the endoscope has to be suspended, the distal end portion of the insertion section has to be separated from the body cavity wall or the like, and the suction operation has to be resumed, thus making treatment burdensome and time-consuming.

(17) Accordingly, the present disclosure is aimed at providing an endoscope and an endoscope system with which suction may be continued without being interrupted even when a channel opening is blocked by a body cavity wall or the like or an opening amount is reduced, and with which time and burden of treatment may be reduced.

(18) In the following, a description will be given of an endoscope system **100** of a present embodiment. As shown in FIG. **1**, the endoscope system **100** includes an endoscope **1**, and a cover member **20** that is a distal end opening/closing cover. Note that a configuration is also possible where the cover member **20** is included in the endoscope **1**.

(19) The endoscope **1** includes an insertion section **2**, an operation section **3**, a universal cord **4**, and an endoscope connector **5**. The insertion section **2** is a long member that can be inserted inside a subject. The insertion section **2** includes a distal end portion **6**, a bending portion **7**, and a flexible tube portion **8**.

(20) The distal end portion **6** is disposed on a distal end of the insertion section **2**. The bending portion **7** is disposed on a proximal end side of the distal end portion **6**. The bending portion **7** is an active bending part and is freely bendable. The flexible tube portion **8** has flexibility, and is disposed on a proximal end side of the bending portion **7**. The flexible tube portion **8** is a passive flexible part, and is connected on a distal end side of the operation section **3**.

(21) A pair of angle knobs **10** for performing bending of the bending portion **7** are provided on the operation section **3**. An air/water feeding button **11**, a suction button **12**, and a plurality of video-related switches **13**, **14** configured to perform operations related to freezing of an endoscope image, releasing of image pickup, and the like are provided on the operation section **3**.

(22) The universal cord **4** is a composite cable extending from a side portion of the operation section **3**. The endoscope connector **5** is provided on an extension end of the universal cord **4**. The endoscope connector **5** is connected to a video processor, not shown, of a camera control unit (CCU) including a light source device.

(23) Moreover, the suction button **12** enables, by being pressed down, suction of foreign substances such as excrement, undigested substances and blood inside a subject by a suction device that is not shown. Note that a treatment instrument insertion channel, described later, is provided from the insertion section **2** to the operation section **3**, and the foreign substances are suctioned through the treatment instrument insertion channel.

(24) The endoscope **1** configured in the above manner is desirably a disposable single-use endoscope that is disposed of after use (after being used once), but may of course be a reusable product that is re-used after being subjected to a disinfecting/sterilizing process.

(25) As shown in FIGS. **2** and **3**, the distal end portion **6** of the insertion section **2** includes a distal end structural member **15** that is made of metal or the like and that is formed into a bullet shape. An observation window **31** and an illumination window **32** are provided in a distal end surface **30** of the distal end structural member **15**.

(26) Photographing light entering from the observation window **31** is photoelectrically converted by an image pickup unit **40** built inside the distal end portion **6**. An image pickup cable **41** extending from the image pickup unit **40** is inserted through the insertion section **2**, the operation section **3**, and the universal cord **4**, up to the endoscope connector **5**.

(27) Illumination light is directed toward the subject from the illumination window **32**. The illumination light is transmitted by a light guide that is not shown. The light guide is inserted through the insertion section **2**, the operation section **3**, and the universal cord **4**, up to the endoscope connector **5**. Note that the illumination light is emitted by a light source that is built inside the video processor.

(28) The endoscope **1** here is configured such that the illumination light emitted from the video processor is transmitted by the light guide, but instead, a light source such as an LED may be provided in the distal end portion **6**.

(29) As shown in FIG. **3**, a bending rubber connection tube **42** is fitted on a proximal end of the distal end structural member **15**. The bending rubber connection tube **42** is connected while being covered by a distal end part of a bending rubber **43**. Note that the bending rubber **43** covers a plurality of bending pieces **44** provided on the bending portion **7** in an integrated manner.

(30) A channel conduit **35** is provided in the distal end structural member **15**. A distal end opening portion **33** of the channel conduit **35** is formed in the distal end surface **30** of the distal end structural member **15**. In other words, the distal end opening portion **33** is an opening on a front side of the channel conduit **35**.

(31) A channel connection tube **45** that communicates with the channel conduit **35** is fitted in the distal end structural member **15**. A distal end part of a treatment instrument insertion channel **46** that functions also as a suction channel is connected to the channel connection tube **45**.

(32) A side surface opening portion **34** is formed in a side surface of the distal end structural member **15**, the side surface opening portion **34** being formed by being cut out up to the channel conduit **35**. In other words, the side surface opening portion **34** is a lateral opening of the channel conduit **35**. The side surface opening portion **34** communicates with the treatment instrument insertion channel **46**. Accordingly, the distal end structural member **15** is provided with the side surface opening portion **34** that is on a lateral side and that communicates with the distal end opening portion **33** on a front side.

(33) A groove **36** is formed on the distal end structural member **15**, the groove **36** being formed around a longitudinal axis on an outer circumferential portion along the side surface opening portion **34**. The groove **36** is a bottomed groove portion that is formed in a recessed manner in a circumferential direction at a position of the side surface opening portion **34** of the distal end structural member **15**.

(34) The cover member **20** is attached to the groove **36**. Note that as shown in FIG. **4**, the cover member **20** is engaged by being fitted to cover the groove **36** from a distal end side of the distal end structural member **15**. The cover member **20** is a disposable single-use cover member that is disposed of after use (after being used once).

(35) As shown in FIGS. **5** and **6**, the cover member **20** is a cylindrical body that is formed from an elastic material such as elastomer or silicone rubber. The cover member **20** includes two slits **21** that are straight and that are formed from positions close to a distal end side to a proximal end.

(36) The two slits **21** are formed to be parallel to each other with a predetermined distance in between in a circumferential direction of the cover member **20**. A gap between the two slits **21** is set smaller than a width range (region) of the side surface opening portion **34** orthogonal to a longitudinal axis of the distal end structural member **15**.

(37) An open/close valve **22** (secondary body) is formed between the two slits **21**, the open/close valve **22** being an opening/closing portion that is arch-shaped in a cross-sectional direction and rectangular in a planar direction. Normally, the open/close valve **22** is retained by an elastic force such that the cover member **20** is maintained in a substantially cylindrical shape.

(38) Note that the open/close valve **22** is arch-shaped in the cross-sectional direction, and is not easily deformed to open outward relative to the distal end portion **6**. Accordingly, unintended deformation of the open/close valve **22** in an outward direction relative to the distal end portion **6** may be prevented.

(39) The cover member **20** is attached by being engaged with the groove **36** of the distal end structural member **15**. The side surface opening portion **34** is thus covered by the cover member **20**.

(40) Note that a depth of the groove **36** and a thickness of the cover member **20** are substantially the same. A length of the groove **36** and a length of the cover member **20** in a longitudinal axis direction are also substantially the same. Accordingly, the cover member **20** is attached in such a way that an outer circumferential surface of the cover member **20** is on a substantially same plane with an outer surface of the distal end structural member **15** with no step between the two.

(41) Note that the cover member **20** is attached to the groove **36** of the distal end structural member **15** in such a way that, in relation to positions around an axis, the two slits are positioned within the width range of the side surface opening portion **34**. Accordingly, the open/close valve **22** formed by the two slits **21** is positioned within the width range of the side surface opening portion **34**.

(42) With the endoscope system **100** configured in the above manner, normally, the open/close valve **22** of the cover member **20** attached to the distal end portion **6** is in a closed state, as shown in FIGS. **2** and **3**. In other words, the cover member **20** is substantially cylindrically shaped, with the open/close valve retained by the elastic force.

(43) Accordingly, the side surface opening portion **34** formed in the distal end structural member **15** of the distal end portion **6** is closed by being entirely covered by the cover member **20**. Accordingly, the channel conduit **35** of the distal end structural member **15** is open only at the distal end opening portion **33**.

(44) When suction is performed in the state described above, the endoscope system **100** suctions foreign substances such as excrement, undigested substances and blood inside the subject from the distal end opening portion **33**. Suction is performed from the distal end opening portion **33**, through the channel conduit **35** and the treatment instrument insertion channel **46**.

(45) Note that because the open/close valve **22** is closed, air and the like inside the body cavity are not suctioned through the side surface opening portion **34** on an outer circumferential side of the distal end portion **6** of the endoscope **1**, and reduction in a suction force may be prevented.

(46) As shown in FIG. **7**, the endoscope system **100** may perform suction while the distal end surface **30** of the distal end portion **6** of the insertion section **2** is in contact with a body cavity wall **101**. The body cavity wall **101** such as a stomach wall sticks to the distal end opening portion **33** due to the suction force. At this time, the distal end opening portion **33** is placed in a completely blocked state or in a state where an opening amount is reduced due to the body cavity wall **101**.

(47) In this state, negative pressure with predetermined magnitude is generated inside the channel conduit **35** of the distal end structural member **15** from the treatment instrument insertion channel **46**. Due to the negative pressure, the open/close valve **22** of the cover member **20** moves in such a way as to be drawn into the channel conduit **35**. In other words, the open/close valve **22** deforms toward an inside of the channel conduit **35**, and the side surface opening portion **34** is placed in an open state. The channel conduit **35** of the distal end structural member **15** is thus opened at the side surface opening portion **34**.

(48) In this state, the endoscope system **100** is able to suction foreign substances such as excrement, undigested substances and blood inside the subject from the side surface opening portion **34**. The suction is performed from the side surface opening portion **34**, through the channel conduit **35** and the treatment instrument insertion channel **46**.

(49) As described above, the endoscope system **100** may continue performing suction without being interrupted even when the distal end opening portion **33** of the distal end portion **6** of the insertion section **2** adheres to the body cavity wall **101** such as a stomach wall or the opening amount is reduced during suction. This is because the open/close valve **22** of the cover member **20** covering the side surface opening portion **34** is opened due to the negative pressure generated inside the channel conduit **35** from the treatment instrument insertion channel **46**.

(50) Note that when the distal end opening portion **33** is blocked by the body cavity wall **101**, the open/close valve **22** moves by being deformed by the suction force of the endoscope **1** and the side

surface opening portion **34** is thus opened. Settings here are appropriately managed based on a length, a shape, a material, a thickness and the like of the two slits **21** formed in the cover member **20**.

(51) According to the description given above, at the time of suction inside a subject by the endoscope **1**, the endoscope system **100** may continue performing suction through the side surface opening portion **34** without being interrupted even when the body cavity wall **101** such as a stomach wall sticks to the distal end portion **6** of the insertion section **2** and the distal end opening portion **33** of the channel opening is blocked or the opening amount is reduced. Accordingly, a suction procedure may be performed with high responsiveness.

(52) Accordingly, with the endoscope system **100**, time and burden of a task of suspending suction by the endoscope **1**, separating the distal end portion **6** of the insertion section **2** from the body cavity wall **101**, and resuming suction may be eliminated. Accordingly, with the endoscope system **100**, a suction treatment may be performed more efficiently than with a conventional configuration.

(53) Note that, depending on the length, the shape, the material, the thickness and the like of the two slits **21** formed in the cover member **20**, an amount of deformation of the open/close valve **22** is set in the following manner, for example. Here, an opening amount of the side surface opening portion **34** that is moved due to deformation of the open/close valve **22** at a time when the distal end opening portion **33** is completely blocked by the body cavity wall **101** (100%) is taken as a reference value (100%).

(54) In a case where the distal end opening portion **33** is blocked 70% by the body cavity wall **101** and the opening amount is 30%, the open/close valve **22** is deformed such that the opening amount of the side surface opening portion **34** is 10% relative to the reference value. The minimum suctioning flow causing deformation of the open/close valve **22** is 30% about the opening amount of the distal end opening portion **33**.

(55) Furthermore, in a case where the distal end opening portion **33** is blocked 80% by the body cavity wall **101** and the opening amount is 20%, the open/close valve **22** is deformed such that the opening amount of the side surface opening portion **34** is 30% relative to the reference value.

(56) Moreover, in a case where the distal end opening portion **33** is blocked 90% by the body cavity wall **101** and the opening amount is 10%, the open/close valve **22** is deformed such that the opening amount of the side surface opening portion **34** is 50% relative to the reference value.

(57) The suctioning flow can include air and liquid, such as blood. Also, the cases described above are merely examples, and various amounts of deformation may be set for the open/close valve **22** depending on the length, the shape, the material, the thickness and the like of the two slits **21** formed in the cover member **20**.

(58) (Modification)

(59) (First Modification)

(60) In the case where suction from a side surface of the distal end portion **6** is not necessary, the endoscope system **100** may cause the open/close valve **22** to be moved outside the region of the side surface opening portion **34**, as shown in FIG. **8**. More specifically, the cover member **20** may be used by being rotated around the axis. In this case, the endoscope **1** performs suction of foreign substances only through the distal end opening portion **33**.

(61) (Second Modification)

(62) As shown in FIG. **9**, two open/close valves **22** may be formed by providing three slits **21** in the cover member **20**.

(63) When the open/close valve **22** is divided into two and a width is reduced, the two open/close valves **22** of the cover member **20** may be easily deformed. Accordingly, the two open/close valves **22** are opened even with small negative pressure that is generated inside the distal end structural member **15** due to a small suction force.

(64) Accordingly, with the endoscope system **100**, foreign substances may be easily suctioned through the side surface opening portion **34** of the distal end portion **6** when the distal end opening

portion **33** is blocked by the body cavity wall **101** and the opening amount is reduced. Accordingly, the endoscope system **100** achieves better responsiveness in relation to suction from the side surface opening portion **34** in a state where the distal end opening portion **33** is blocked.

(65) Moreover, in a state where one of the two open/close valves **22** is in contact with the body cavity wall **101**, the other open/close valve **22** is opened/closed. In other words, even when the distal end opening portion **33** and one of the open/close valves **22** blocking the side surface opening portion **34** adhere to the body cavity wall **101** of a living body, the other open/close valve **22** is deformed to open the side surface opening portion **34**. Accordingly, with the endoscope system **100**, suction by the endoscope **1** may be more smoothly performed. As described above, the two open/close valves **22** may be separately opened/closed depending on the situation.

(66) (Third Modification)

(67) As shown in FIG. **10**, the two slits **21** of the cover member **20** may be obliquely formed such that a gap of the two slits **21** is reduced toward a distal end side. In other words, the open/close valve **22** having a trapezoidal shape where the distal end side is a top side and a proximal end side is a bottom side may be formed. In other words, the two slits **21** are formed to be not parallel to each other along a longitudinal axis of the cover member **20**.

(68) Accordingly, because a width on the distal end side is small, the open/close valve **22** is easily deformed. Accordingly, the endoscope system **100** achieves better responsiveness in relation to suction from the side surface opening portion **34** in a state where the distal end opening portion **33** is blocked.

(69) (Fourth Modification)

(70) As shown in FIG. **11**, the two slits **21** of the cover member **20** may be obliquely formed such that the gap the two slits **21** is reduced toward the proximal end side. In other words, the open/close valve **22** having a trapezoidal shape where the distal end side is the bottom side and the proximal end side is the top side may be formed. In other words, the two slits **21** are formed to be not parallel to each other along the longitudinal axis of the cover member **20**.

(71) Accordingly, because the width on the distal end side is great, the open/close valve **22** is not easily deformed on the distal end side but is easily deformed on the proximal end side.

Accordingly, the negative pressure inside the distal end portion **6** of the endoscope **1** changes depending on an amount of reduction in the opening amount of the blocked distal end opening portion **33**. Therefore, the width of the open/close valve **22** is increased toward a distal end, and a position at which the open/close valve **22** deforms changes according to a change in the negative pressure.

(72) With the endoscope system **100**, when the open/close valve **22** is not easily deformed on the distal end side but is easily deformed on the proximal end side, the opening amount of the side surface opening portion **34** by the open/close valve **22** may be easily controlled according to magnitude of the negative pressure inside the distal end portion **6**. As a result, the open/close valve **22** of the present modification is suitable in a case where the suction force of the side surface opening portion **34** is desired to be controlled.

(73) (Fifth Modification)

(74) As shown in FIG. **12**, an open/close valve **50** having a thin plate shape may be provided at the distal end structural member **15** constituting the distal end portion **6** of the endoscope **1**, without providing the cover member **20**.

(75) The open/close valve **50** is made of synthetic resin, and a groove or a thin-filmed V-shaped groove **51** is formed in a width direction on an inner surface on the distal end side. The open/close valve **50** is disposed to block the side surface opening portion **34** of the distal end structural member **15**.

(76) As shown in FIG. **13**, when the distal end opening portion **33** is blocked by the body cavity wall **101** at the time of suction and the opening amount is reduced, the open/close valve **50** moves by being deformed toward an inside of the channel conduit **35** around the V-shaped groove **51**. In

other words, the open/close valve **50** moves in such a way as to be drawn into the channel conduit **35** by the negative pressure with predetermined magnitude that is generated inside the channel conduit **35** of the distal end structural member **15** from the treatment instrument insertion channel **46**.

(77) The side surface opening portion **34** is thereby placed in an open state. In this state, the endoscope **1** is able to suction foreign substances such as excrement, undigested substances and blood inside the subject from the side surface opening portion **34**, through the channel conduit **35** and the treatment instrument insertion channel **46**.

(78) The present disclosure is not limited to the embodiment described above, and changes may be made as appropriate within the gist or the idea of the disclosure that can be read from the claims, the entire specification, and the drawings.

Claims

1. An endoscope, comprising: an insertion section including a distal end portion; a distal end opening in a distal end surface of the distal end portion; a side opening in a side surface of the distal end portion; a suction channel located in the insertion section and extending proximally from the distal end portion, wherein the suction channel is in communication with the distal end opening and with the side opening; and a valve switchable between an open state and a closed state, wherein, when a first suctioning flow through the distal end opening is less than a minimum suctioning flow, the valve is in the open state and a second suctioning flow through the side opening is present, and wherein, when the first suctioning flow through the distal end opening is more than the minimum suctioning flow, the valve is in the closed state and the second suctioning flow through the side opening is absent.
2. The endoscope according to claim 1, wherein the first suctioning flow is a first suctioning liquid flow and the second suctioning flow is a second suctioning liquid flow.
3. The endoscope according to claim 1, wherein the minimum suctioning flow is 30% of a reference suctioning flow, and wherein the reference suctioning flow is a suctioning flow through the distal end opening in a fully open state.
4. The endoscope according to claim 3, wherein, in the open state, the first suctioning flow and the second suctioning flow is less than the reference suctioning flow.
5. The endoscope according to claim 1, wherein the valve is switchable from the closed state to the open state by a negative pressure generated inside the suction channel.
6. The endoscope according to claim 1, further comprising a distal cover located in the distal end portion, wherein a surface of the distal cover covers the side opening, wherein the valve is located in the surface of the distal cover, wherein the distal cover is rotatable relative to the distal end portion around a longitudinal axis of the distal end portion, between a first position and a second position, wherein, in the first position, the valve is aligned with the side opening, and wherein, in the second position, the valve is unaligned with the side opening.
7. The endoscope according to claim 6, wherein the valve includes a main body and a secondary body, and wherein the secondary body moves toward an interior space of the distal end portion by a negative pressure.
8. The endoscope according to claim 6, wherein the valve is formed by a plurality of slits provided in the distal cover.
9. The endoscope according to claim 8, wherein the plurality of slits are formed along the longitudinal axis.
10. The endoscope according to claim 8, wherein two adjacent slits of the plurality of slits are separated along the surface of the distal cover by a gap, and wherein a distance of the gap at a distal end of the slits is less than a distance of the gap at a proximal end of the slits.
11. The endoscope according to claim 8, wherein two adjacent slits of the plurality of slits are

separated along the surface of the distal cover by a gap, and wherein a distance of the gap at a distal end of the slits is more than a distance of the gap at a proximal end of the slits.

12. The endoscope according to claim 1, wherein the valve is formed from an elastic material.

13. The endoscope according to claim 1, wherein the distal end surface of the distal end portion includes a first distal end surface section that is non-perpendicular to a longitudinal axis of the distal end portion, and wherein at least a majority portion of the distal end opening is in the first distal end surface section.

14. The endoscope according to claim 13, wherein the distal end surface of the distal end portion includes a second distal end surface section that is perpendicular to a longitudinal axis of the distal end portion, and wherein at least one of an observation window and an illumination window are located in the second distal end surface section.

15. An endoscope system, comprising: an endoscope, wherein the endoscope includes: an insertion section including a distal end portion, a first opening in a distal end surface of the distal end portion, a second opening in a side surface of the distal end portion, and a suction channel located in the insertion section and extending proximally from the distal end portion, wherein the suction channel is in communication with the distal end opening and with the side opening; and a distal cover attached to the distal end portion, wherein the distal cover includes: a valve switchable between open state and closed state, wherein, when a first suctioning flow through the distal end opening is less than a minimum suctioning flow, the valve is in the open state and a second suctioning flow through the side opening is present, and wherein, when the first suctioning flow through the distal end opening is more than the minimum suctioning flow, the valve is in the closed state and the second suctioning flow through the side opening is absent.

16. The endoscope according to claim 15, wherein the first suctioning flow is a first suctioning liquid flow and the second suctioning flow is a second suctioning liquid flow.

17. The endoscope system according to claim 15, wherein the minimum suctioning flow is 30% of a reference suctioning flow, and wherein the reference suctioning flow is a suctioning flow through the distal end opening in a fully open state.

18. The endoscope system according to claim 17, wherein, in the open state, the first suctioning flow and the second suctioning flow is less than the reference suctioning flow.

19. The endoscope system according to claim 15, wherein the valve is switchable from the closed state to the open state by a negative pressure generated inside the suction channel.

20. The endoscope system according to claim 15, wherein the valve is formed by a plurality of slits provided in the distal cover.
